

KSHITIZ GOYAL

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Education

Vellore Institute of Technology (VIT) | CGPA: 8.43

Bachelor of Technology in Computer Science (AI & ML)

Bhopal, Madhya Pradesh, India

Sept 2023 – July 2027

Skills

Programming Languages: Python, SQL, C++

Statistical Modeling & Forecasting: XGBoost, Prophet, Auto-ARIMA, Time-Series Feature Engineering, Chronological Cross-Validation, Hypothesis Testing

Operations Research & Optimization: Safety Stock Modeling, Reorder Point (ROP) Optimization, Demand & Lead-Time Variance Analysis

Data Engineering & Cloud: Apache Spark (PySpark), AWS (S3, Glue, Athena), Apache Airflow, PostgreSQL, SQLite, Terraform (IaC), Docker

Analytics & BI: Power BI, Streamlit, Plotly, Pandas, Great Expectations

Machine Learning: Scikit-Learn, PyTorch, LLM API Integration, Prompt Engineering

Projects

Retail Demand Forecasting & Inventory Optimization Platform | *Python, SQL, Power BI, Streamlit, XGBoost* | [Github](#) | [Demo](#)

- Engineered an automated end-to-end demand forecasting pipeline processing 3M+ retail transactions across 54 stores and 33 product categories, achieving an 8.32% test MAPE (54.3% error reduction over ARIMA baseline).
- Benchmarked 3 model paradigms (XGBoost, Prophet, Auto-ARIMA) using chronological out-of-time cross-validation; extracted 27 time-series features (lags t-1 to t-30, rolling statistics, and bi-monthly payday cycles).
- Formulated an operations research replenishment engine applying statistical Safety Stock ($Z = 1.65$, 95% service level) and dynamic Reorder Points (ROP), modeling both daily demand volatility (σ_d) and supplier lead-time variance (σ_L).
- Architected a normalized SQLite star schema serving layer (12K+ records) powering an executive Power BI reporting suite and an interactive Streamlit web dashboard, pinpointing \$576 in imminent lost sales exposure across 100 SKU segments.

AlphaLake: Financial Market Data Lakehouse & Analytics Platform | *Python, PySpark, AWS, Airflow* | [Github](#) | [Demo](#)

- Designed a Medallion Data Lakehouse (Bronze, Silver, Gold) to ingest and process high-throughput financial market and orderflow event data, using SHA-256 integrity checksums and trade_date partitioning.
- Built automated data-quality validation pipelines with Great Expectations v1.x, including quarantine routing for anomalous records and schema-drift detection to keep invalid data out of downstream layers.
- Implemented distributed PySpark window transformations to compute rolling technical indicators, achieving a 3.05x speedup over an equivalent Pandas baseline.
- Provisioned the underlying AWS analytics infrastructure with Terraform (IaC) and orchestrated scheduled workflows using Apache Airflow DAGs.
- Developed a 6-page interactive Streamlit & Plotly analytics portal to visualize market performance and pipeline health telemetry.

Achievements & Certifications

Hackathon Finalist (Innovatez GenZ Hackathon 2026, Zentiti): Ranked among the top 11 teams out of 600+ registrations nationwide for pitching SupplyGuard, an AI-powered Supplier Risk Monitoring platform.

MPOnline AI/ML Internship Certificate: Completed a certified internship program in Artificial Intelligence & Machine Learning. [Certificate](#)

Applied Machine Learning, University of Michigan: Certified in applied machine learning techniques and model development. [Certificate](#)